

RUN JANNA

M.Eng. Student, Bio-inspired Robotics

CONTACT

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Nonthaburi, Thailand.

PERSONAL PROFILE

I am a robotics researcher working on neural control for legged and amphibious robots. I combine bio-inspired control, machine learning, embedded systems and physics simulation (MuJoCo, CoppeliaSim) to help robots sense their environment and adapt how they move.

SKILLS

Robotics and Automation Software skills

- C/C++ Programming
- Python Programming
- STM32Cube IDE
- Embedded system (ESP32, STM32)
- Robot Operating Systems (ROS2)
- Robot Simulation (CoppeliaSim, MuJoCo)

CAD Software

- PCB Designing (EasyEDA)
- 3D Modeling (SolidWorks, Onshape)

Soft skills

- Leadership
- Project Management
- Collaboration
- Willingness to learn
- Analytical skills

EXTRA-CURRICULAR

Volunteer Experience

- AI Academy at Wangchan Valley, 2023
  - Assisted as staff member in Robotics Camp.
- The 1st International Symposium on Physical Artificial Intelligence and Robotics, 2025
  - Media Operation Chair

AWARD

- IEEE RAS Member Support Program Travel Grant ($1,500 USD) for the 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)

LANGUAGES

- Thai (Native)
- English (Intermediate)

Profile website
(runjanna26.github.io/run-janna)

EXPERIENCES

***2020 to 2021* | Electronic Engineering Intern**
BRAINWAVE GROUP Co., Ltd., Thailand

- Developed IoT software and hardware and mobile apps for coding education.

***2022 to 2025* | Robotics Researcher**
Vidyasirimedhi Institute of Science and Technology (VISTEC), Information Science & Technology (IST), BRAIN Lab, Thailand

- (Industrial Project) Efficiency Improvement through fired Heater/Furnace Cleaning (REFINE) with Thaioil company.
- (Industrial Project) Advanced Hybrid Legged-wheeled Robot for Internal Vessel Inspection (HERO) Project with PTTEP company.
- (Research) Online Adaptive Impedance Control for Lower-limb Exoskeletons.
- (Research Project Manager) DAVE: Diving beetle-inspired Amphibious robot for Versatile Environments.

CERTIFICATION & PATENT

***2023* | Personnel Qualification and Certification in Nondestructive Testing (NDT)**

- Magnetic Particle Testing (MT) Level II Certification (Yoke Technique).

***2024* | Personnel Qualification and Certification in Confined Space Safety Training**

***2025* | "Wheel-legged Robot", Innovation Patent, PCT International Application**

- No. PCT/IB2025/056485

EDUCATION

***2015 to 2018* | Cert.in Tech.Ed | KING MONGKUT'S UNIVERSITY OF TECHNOLOGY OF NORTH BANGKOK (KMUTNB) | Thai-German Pre-Electrical Engineer School (PEE)**

- Major in Electrical and Electronics, *GPAX 3.69*.

***2018 to 2022* | B.Eng. | KING MONGKUT'S UNIVERSITY OF TECHNOLOGY OF NORTH BANGKOK (KMUTNB) | Department of Electrical and Computer Engineering (ECE)**

- Major in Electrical Engineering (Control), *GPAX 3.57*.

***Apr. 2021 to Jul. 2021* | MULTIMEDIA UNIVERSITY (MMU), MALAYSIA | Department of Electrical Engineering (EE)**

- Under the Student Mobility Programme (Online) between the Multimedia University and King Mongkut's University of Technology North Bangkok, *GPAX 3.92.*

***2025 to Present* | M.Eng. | Vidyasirimedhi Institute of Science and Technology (VISTEC), Information Science & Technology (IST)**

- Major in Bio-Inspired Robotics & Neural Engineering (BRAIN-Lab), GPAX 4.00.

THESIS AND RESEARCH

Publications

- An Implementation of BLDC Control System for Virtual Haptic, the 45th Electrical Engineering Conference (EECON 2022).
- Hybrid Omnidirectional Wheeled Climbing Robot with an Electromagnet for Inspection, 26th International Conference on Climbing and Walking Robots (CLAWAR 2023).
- Adaptable Segmented Straps with Anti-slip Pads for Modular Assistive Knee Exoskeleton Enhancement, 27th International Conference on Climbing and Walking Robots (CLAWAR 2024).
- Online Adaptive Impedance Control with Gravity Compensation for an Interactive Lower-Limb Exoskeleton, 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024, CORE A).
- A Framework for Online Human-Exoskeleton Activity Classification and Gait-Lab Control: Towards Automated Gait Rehabilitation, 2025 International Symposium on Adaptive Motion of Animals and Machines (AMAM 2025).
- LITHE-joint: Variable Stiffness Compliant Spherical Contact Joint in an Under-Actuated System, 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025, CORE A).
- Automated Weld Seam Detection and 3D Map Integration for Industrial Vessel Inspection Using Depth Cameras and 3D LiDAR, 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026, CORE A).

In-progress Publications

- A Multi-Axis Compliant Helical Joint for Embodied Adaptable Stiffness and Environment-Driven Interaction (T-RO 2026, Revise and Resubmit).
- A Modular Hybrid Legged-Wheeled Inspection Robot for Multi-Modal Non-Destructive Testing of Steel Assets (ICRA 2027, Submitted).
- Drag-Area Asymmetrical Coupled Origami-Based Paddle for Underwater Legged Robots (ICRA 2027, Submitted).
- A Teleoperated Mobile Robotic System for Furnace Tube Descaling (ICRA 2027, Submitted).
- Compliance-Informed Lightweight Terrain Classification for Multi-Terrain Amphibious Robot Locomotion and Autonomous Gait Transitions (ICRA 2027, Submitted).